

# Customer Shopping Behavior Analysis

3,900 transactions • 18 features • Strategic insights across demographics, products & subscriptions



# Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Values

Review Rating column

\$59.76

Avg Purchase

Key features: demographics, purchase details, shopping behavior, subscription status

# Data Preparation in Python

## → Cleaned & Standardized

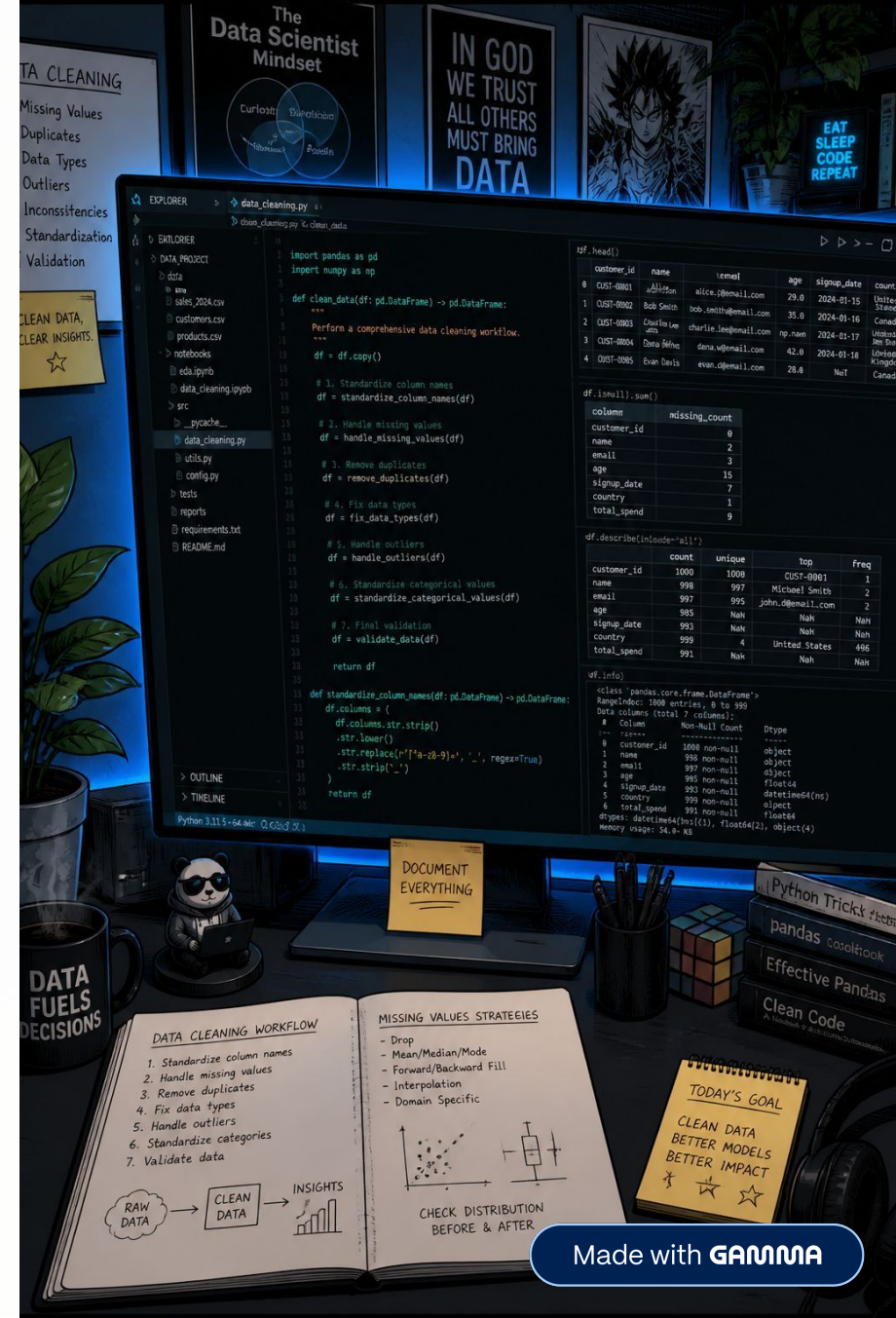
Snake\_case columns, median imputation for Review Rating

## → Feature Engineering

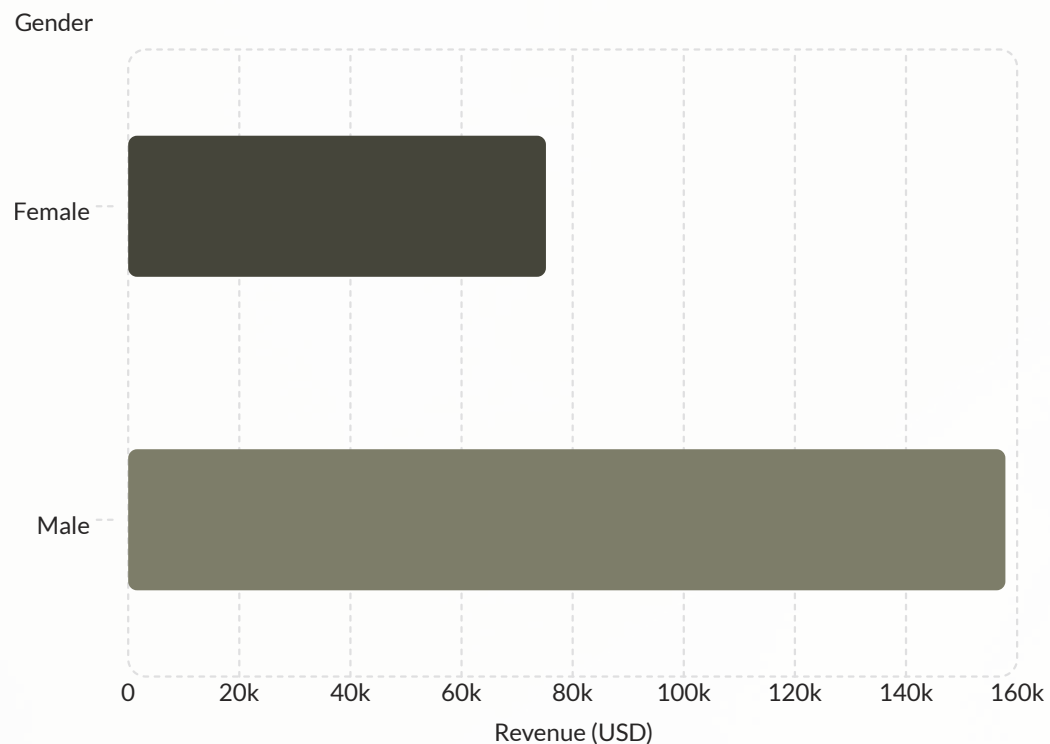
age\_group bins, purchase\_frequency\_days

## → Database Loaded

Cleaned DataFrame exported to PostgreSQL



# Revenue by Gender



Male customers drive 2× more revenue

Male: **\$157,890** • Female: **\$75,191**

Male customers represent 68% of total transactions (2,652 vs. female buyers)

# Top Products & Shipping Insights

Top 5 by Review Rating

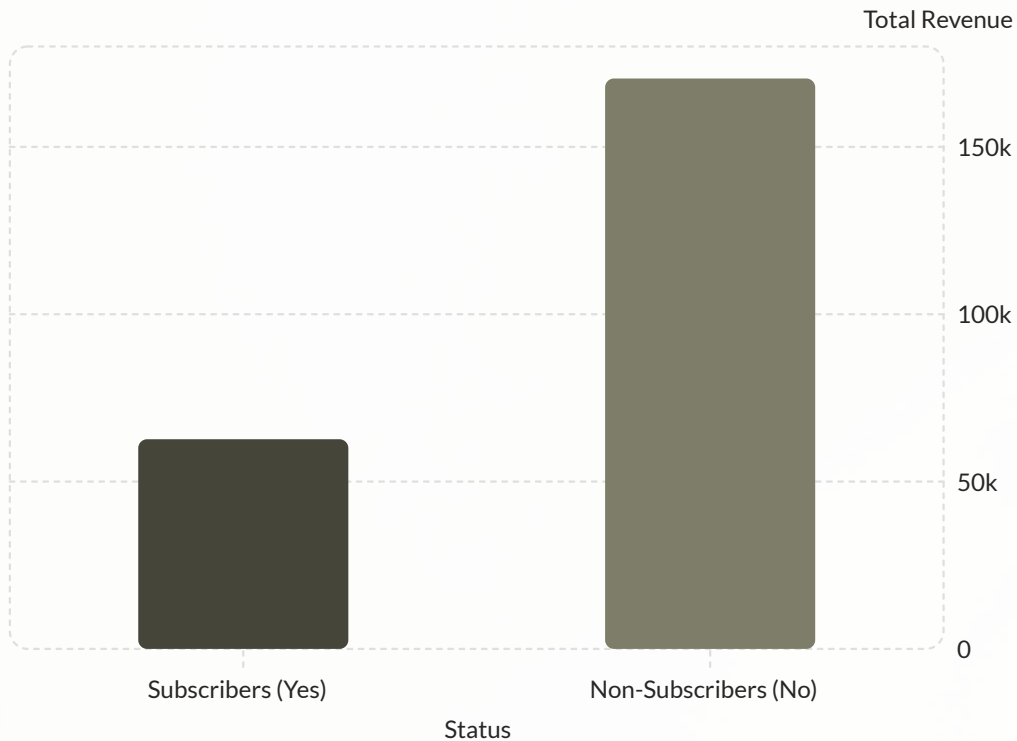
| Product | Avg Rating |
|---------|------------|
| Gloves  | 3.86       |
| Sandals | 3.84       |
| Boots   | 3.82       |
| Hat     | 3.80       |
| Skirt   | 3.78       |

Shipping Type — Avg Spend

|          |         |
|----------|---------|
| Standard | Express |
| \$58.46  | \$60.48 |

Express users spend slightly more per order

# Subscribers vs. Non-Subscribers

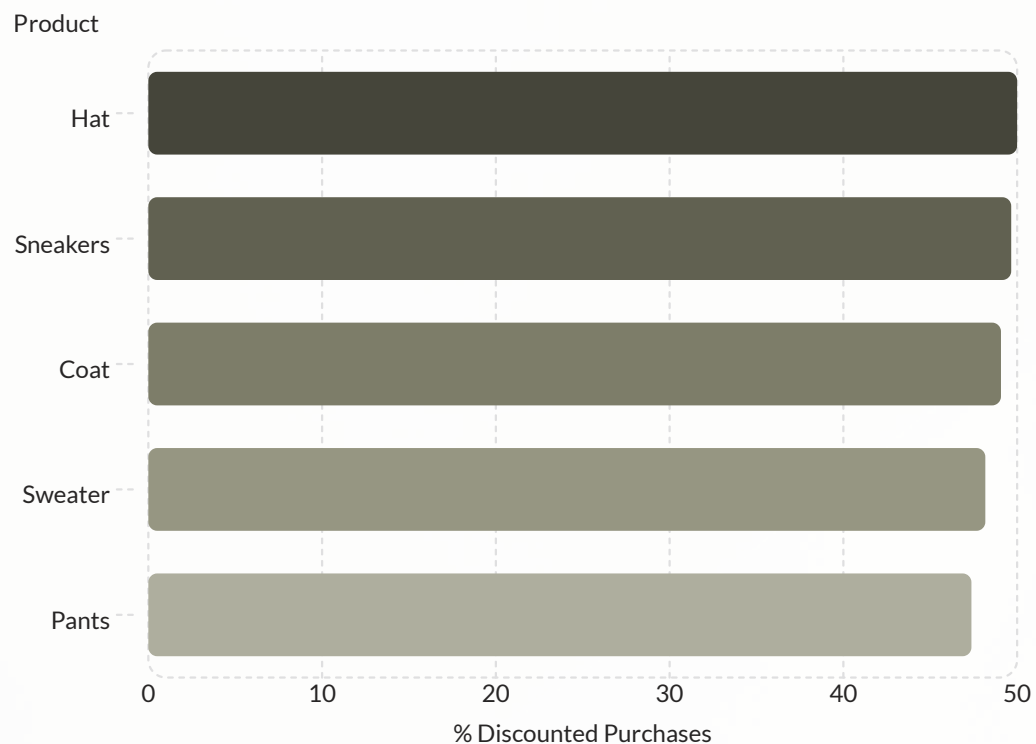


## Key Findings

- Only **27%** are subscribers (1,053 customers)
- Non-subscribers generate **73%** of revenue
- Avg spend nearly identical: \$59.49 vs. \$59.87

📌 Subscribers spend similarly — strong upsell opportunity

# Discount-Dependent Products



~50% of sales rely on discounts

Top 5 products all show ~47–50% discount rates — review policy to protect margins

**839 customers** used discounts and still spent above average purchase amount



# Customer Segmentation

Loyal

3,116 customers

Returning

701 customers

New

83 customers

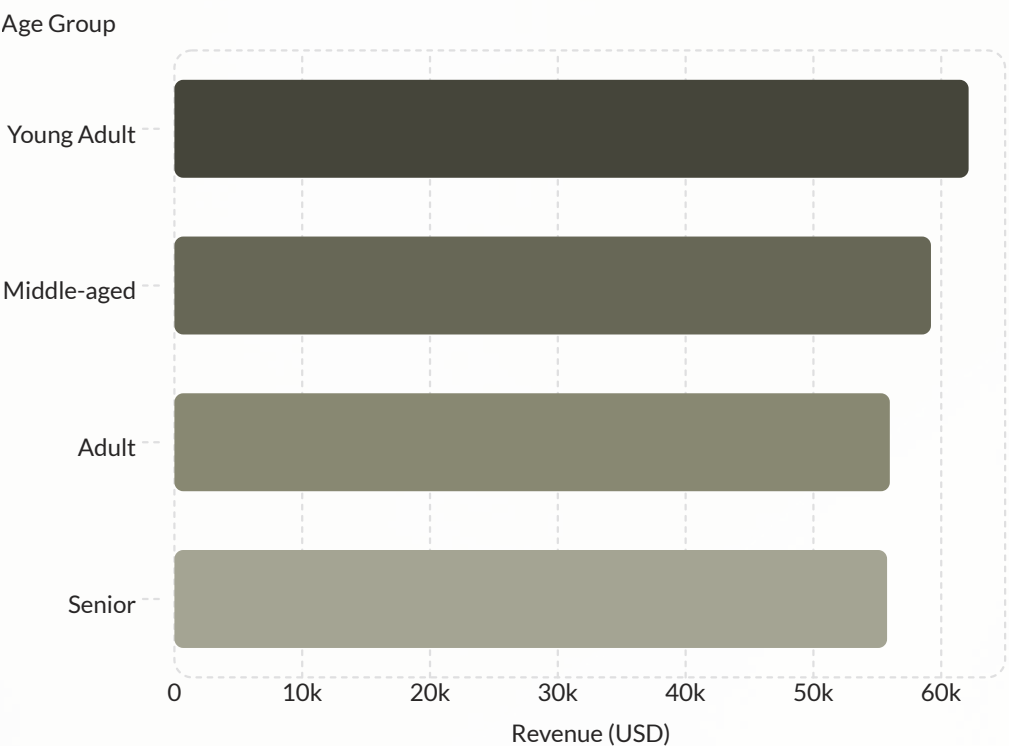
80% of customers are Loyal segment. 958 repeat buyers (>5 purchases) have subscriptions vs. 2,518 without.



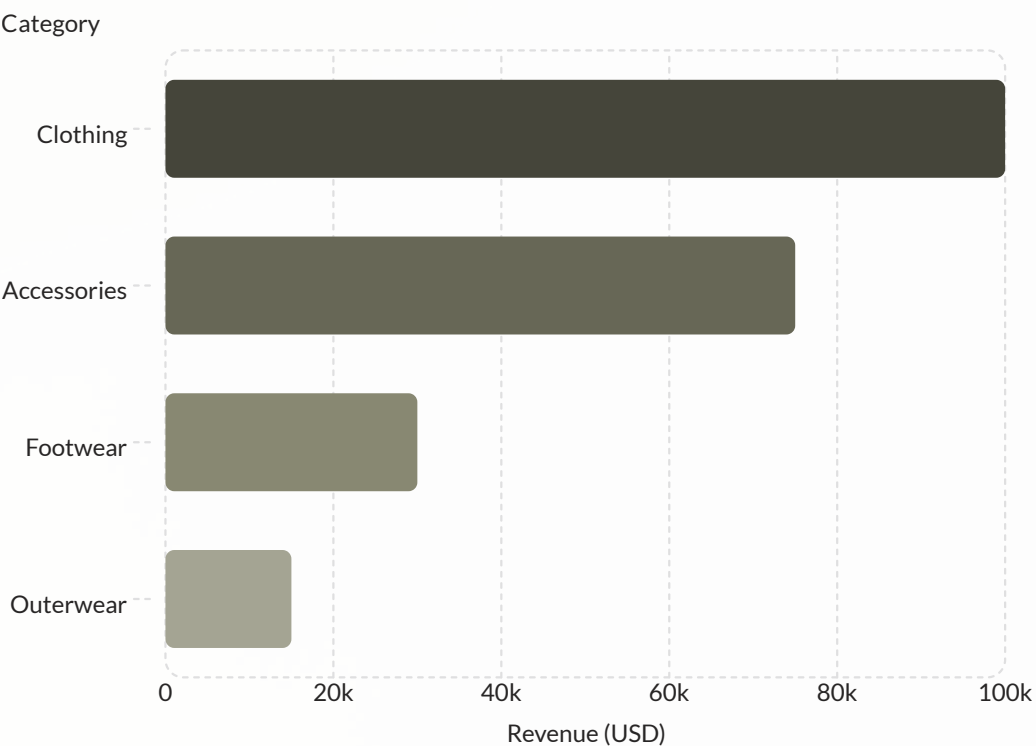


# Revenue by Age Group & Category

Revenue by Age Group



Revenue by Category



Young Adults lead revenue. Clothing dominates at \$100K – 4× Outerwear.



# Business Recommendations



## Boost Subscriptions

Promote exclusive subscriber benefits



## Loyalty Programs

Reward repeat buyers → Loyal segment



## Review Discount Policy

Balance sales lift with margin control



## Targeted Marketing

Focus on Young Adults & express-shipping users